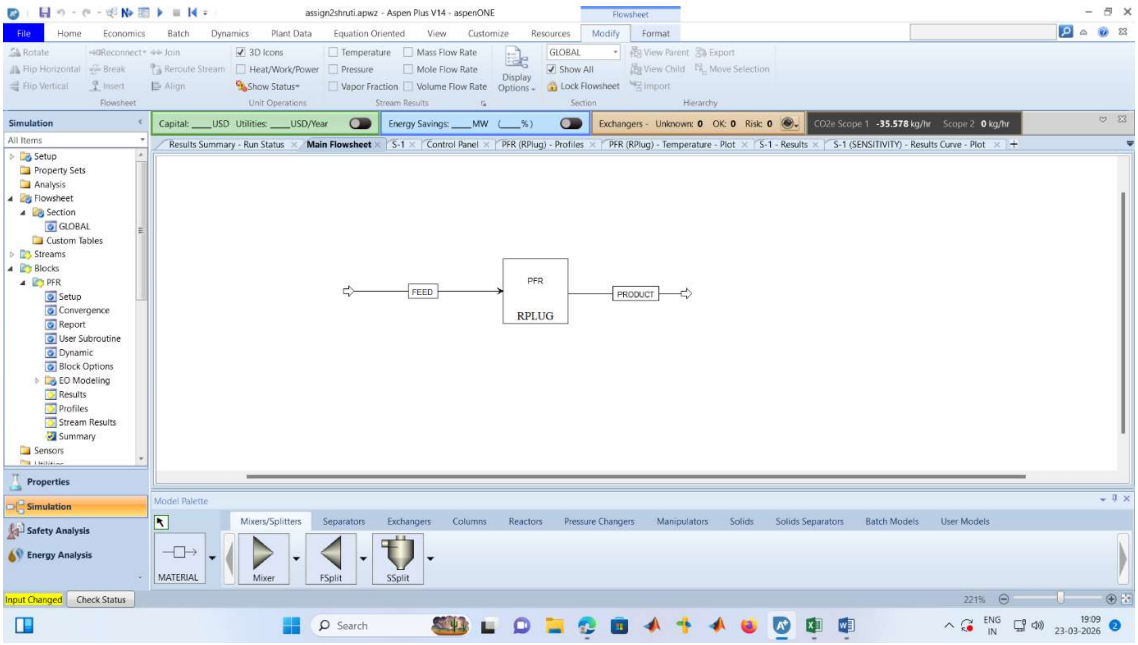
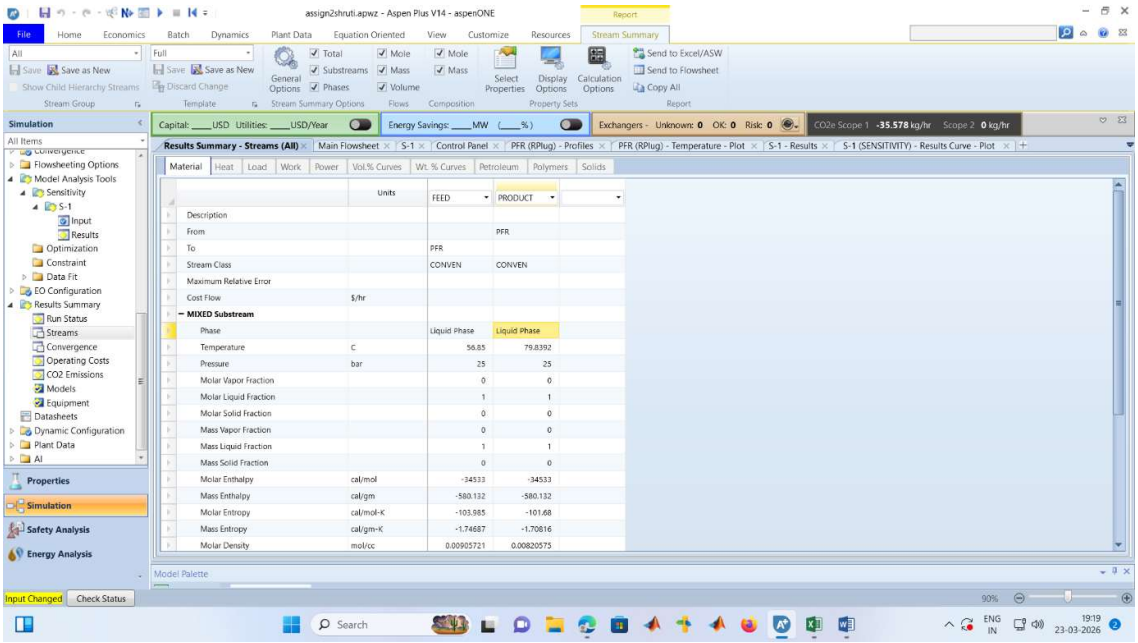
ASSIGNMENT-2
SHRUTI MISHRA (230107070)



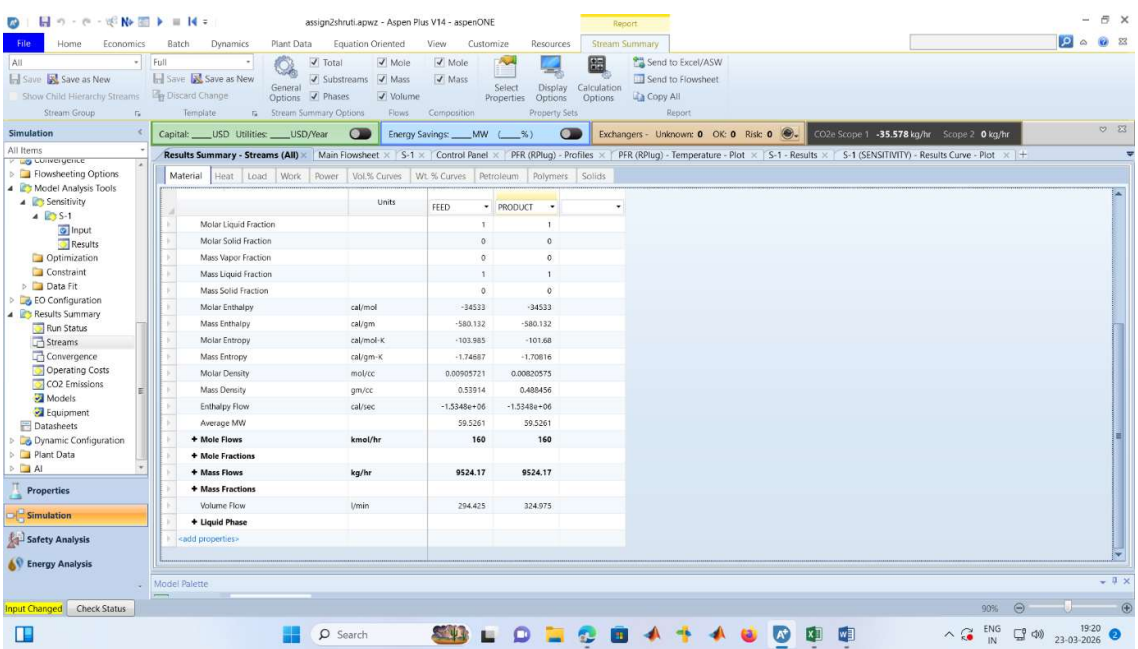
FLOWSHEET OF PFR



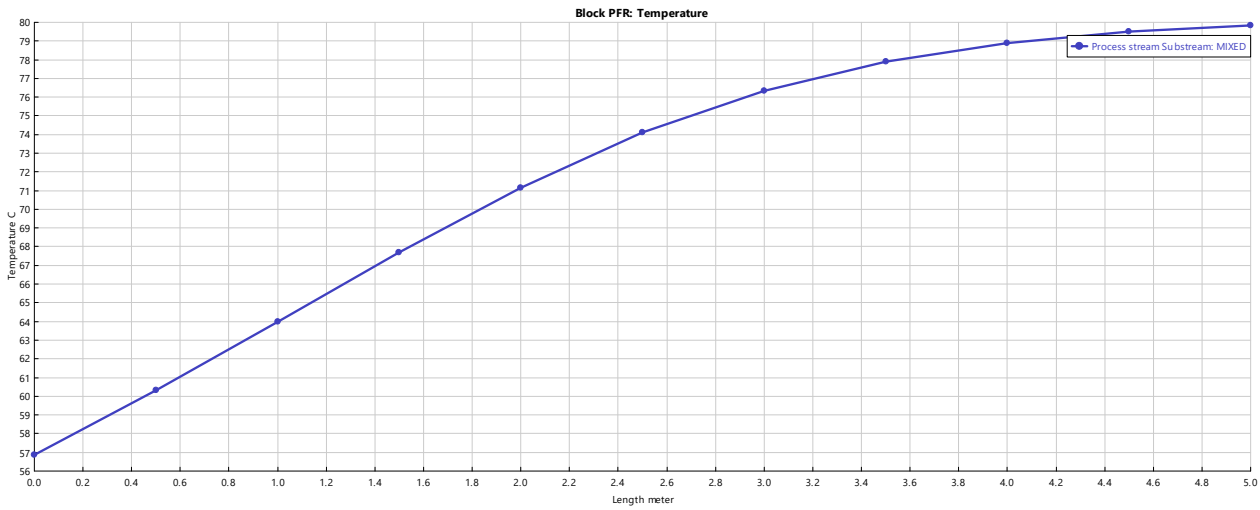
LENGTH OF PFR = 5m

DIAMETER OF PFR = 1m

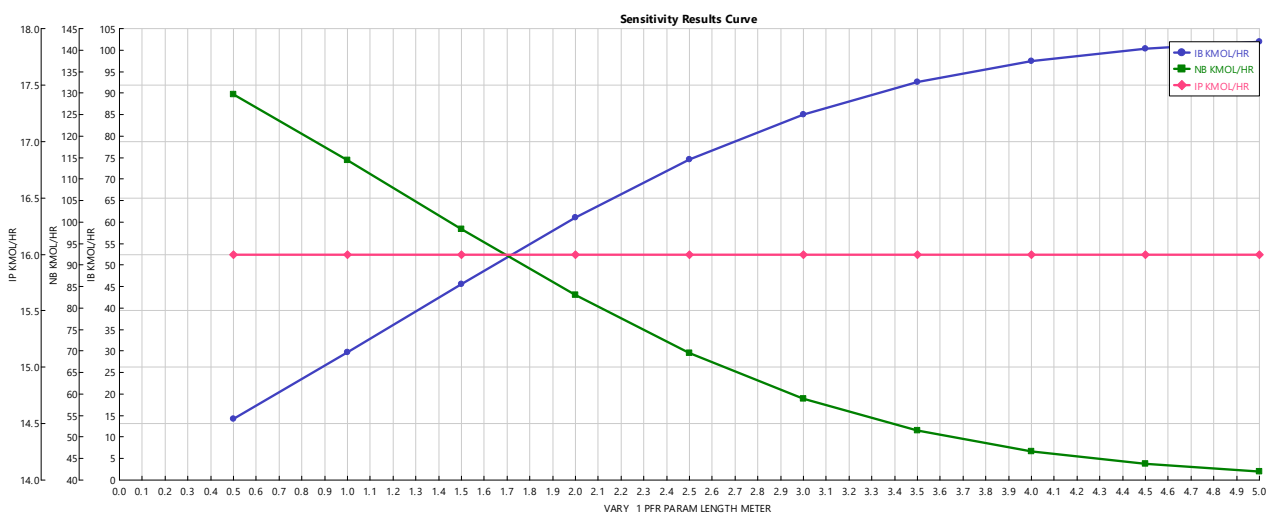
VOLUME OF PFR REQUIRED TO ACHIEVE 70% CONVERSION =
PI*D^2*L/4 = 3.925 m^3

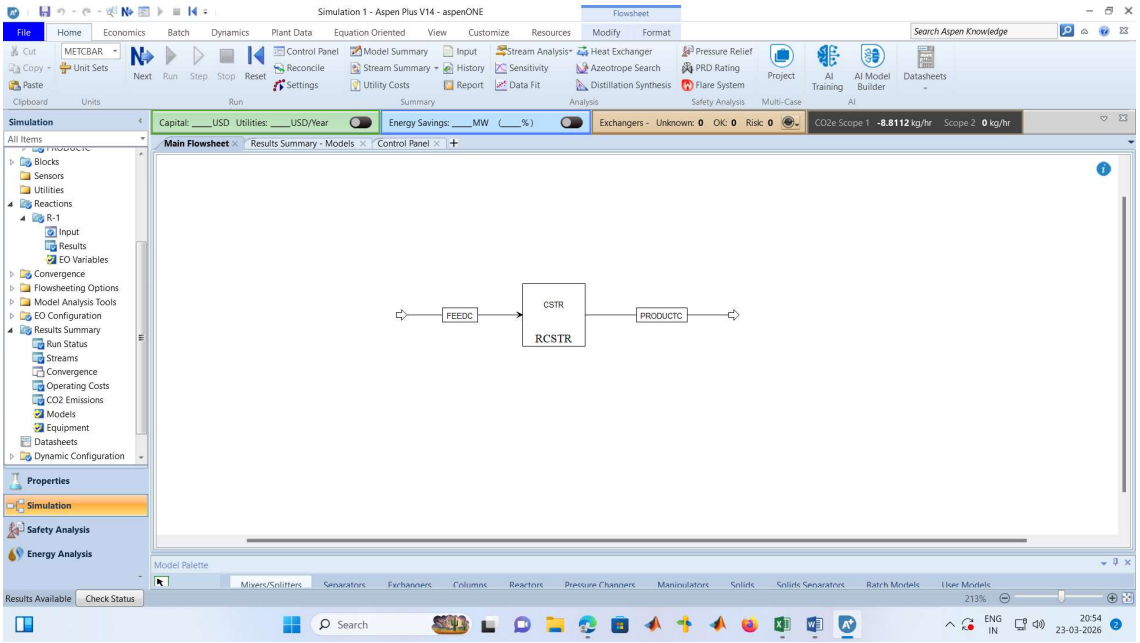


TEMPERATURE VS LENGTH OF REACTOR

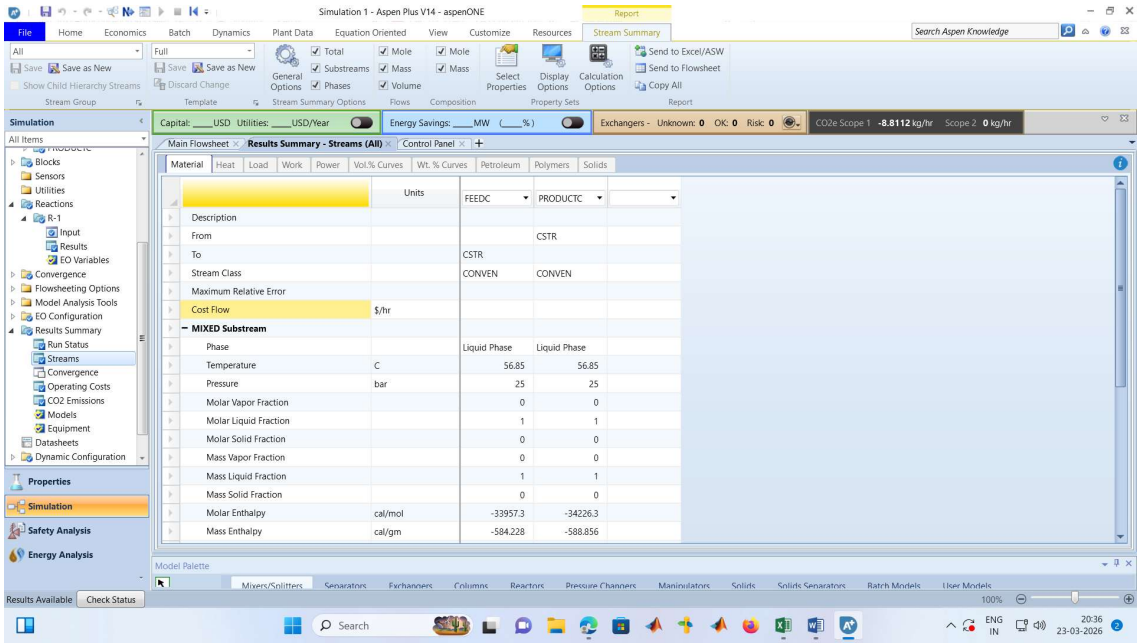


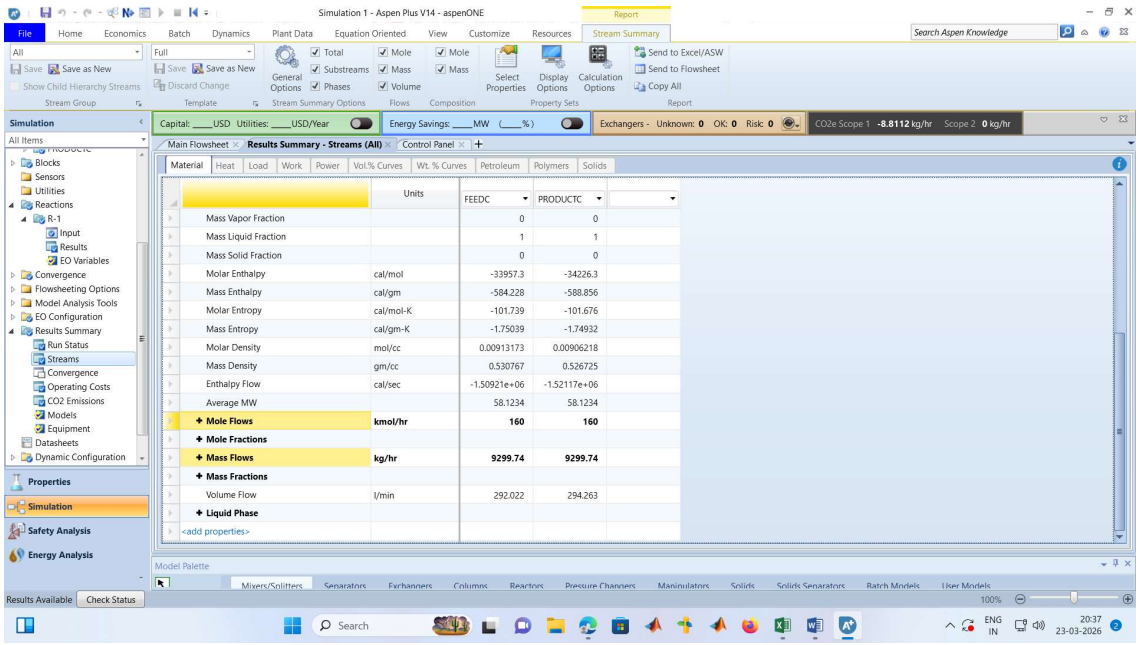
MOLAR COMPOSITION VS LENGTH OF REACTOR

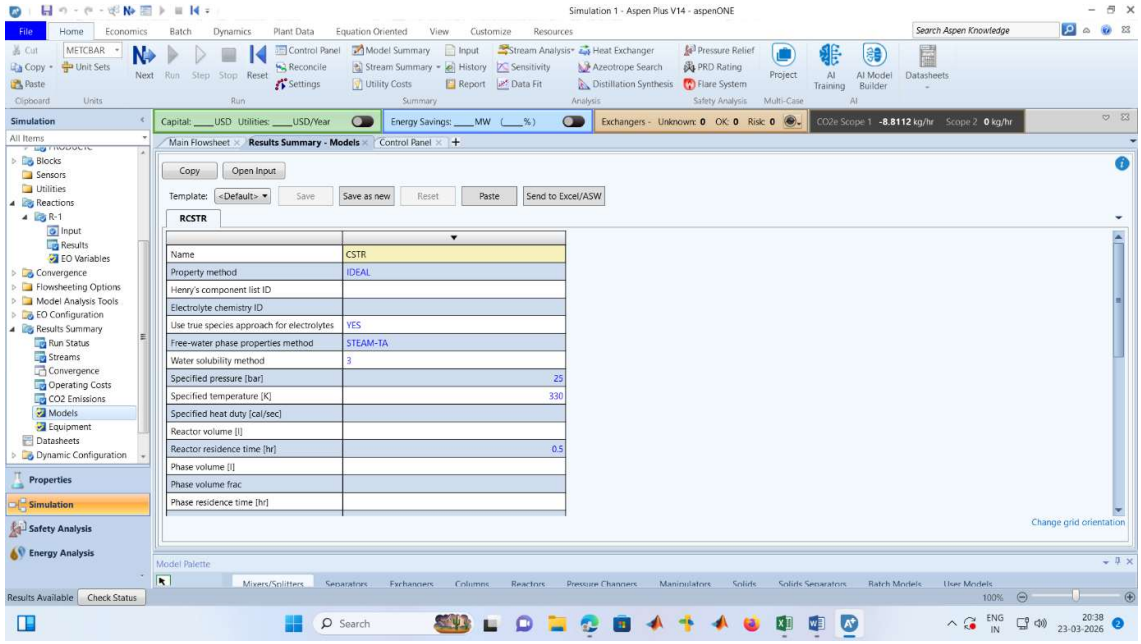




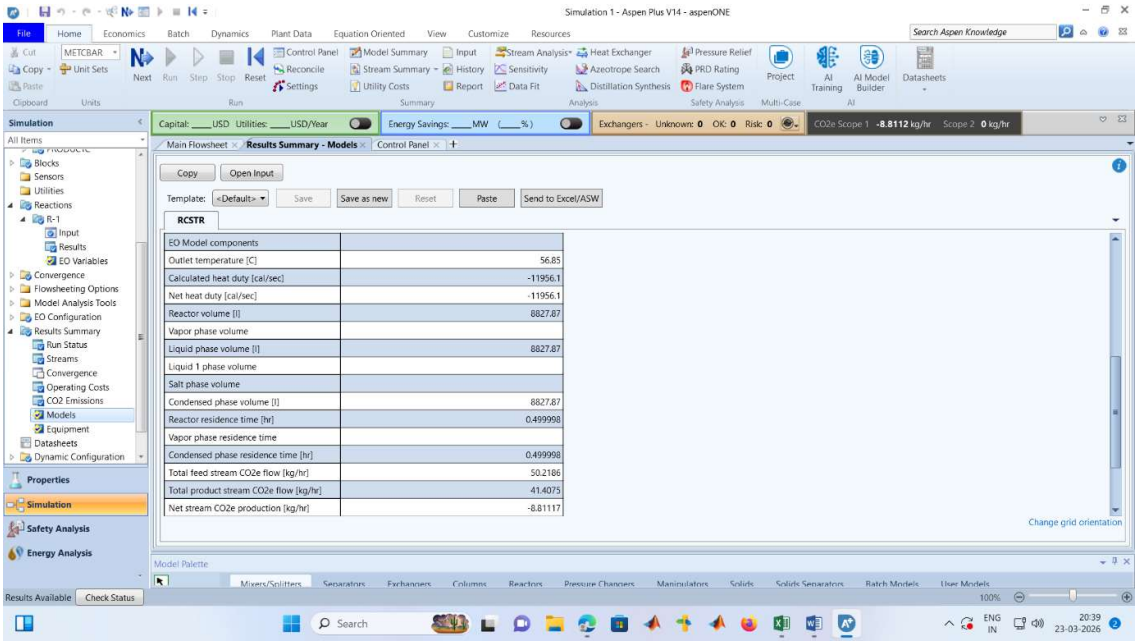
FLOWSHEET OF CSTR



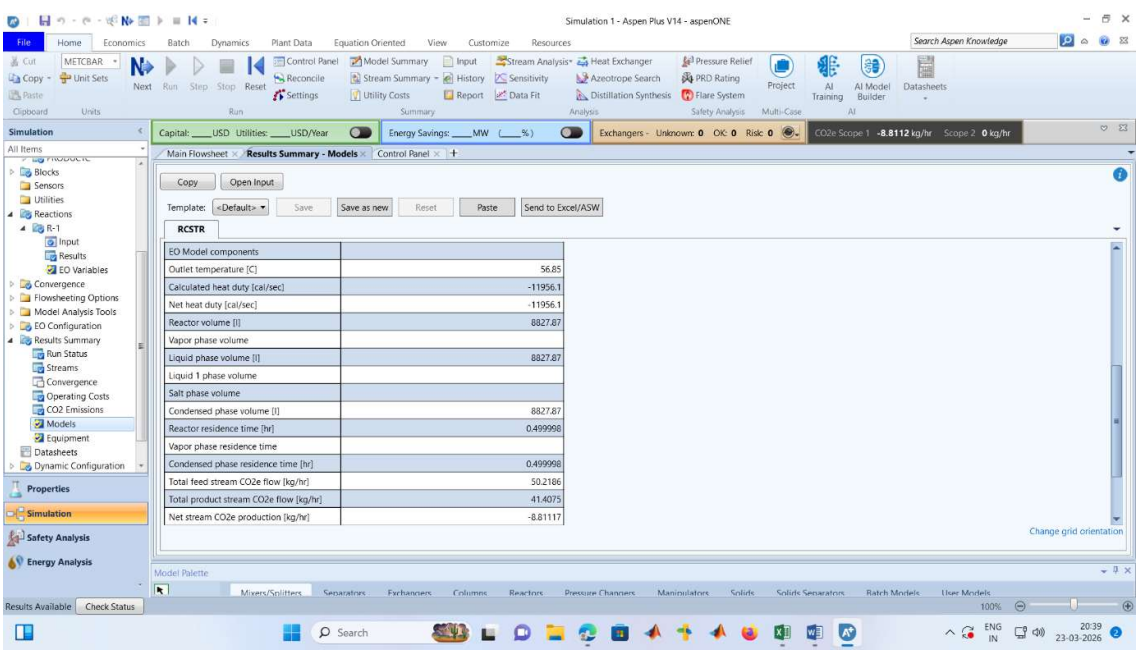

Simulation 1 - Aspen Plus V14 - aspenONE
Main Flowsheet
Results Summary - Streams (All)
Control Panel
Material
Heat
Load
Work
Power
Vol.% Curves
Wt. % Curves
Petroleum
Polymers
Solids
Units
FEEDC
PRODUCTC
Mass Vapor Fraction 0 0
Mass Liquid Fraction 1 1
Mass Solid Fraction 0 0
Molar Enthalpy cal/mol -33957.3 -34226.3
Mass Enthalpy cal/gm -584.228 -588.856
Molar Entropy cal/mol-K -101.739 -101.676
Mass Entropy cal/gm-K -1.75039 -1.74932
Molar Density mol/cc 0.00913173 0.00906218
Mass Density gm/cc 0.530767 0.526725
Enthalpy Flow cal/sec -1.50921e+06 -1.52117e+06
Average MW 58.1234 58.1234
Mole Flows kmol/hr 160 160
Mole Fractions
Mass Flows kg/hr 9299.74 9299.74
Mass Fractions
Volume Flow l/min 292.022 294.263
Liquid Phase
<add properties>


Simulation 1 - Aspen Plus V14 - aspenONE
Main Flowsheet
Results Summary - Models
Control Panel
Copy
Open Input
Template: <Default>
Save
Save as new
Reset
Paste
Send to Excel/ASW
RCSTR
Name CSTR
Property method IDEAL
Henry's component list ID
Electrolyte chemistry ID
Use true species approach for electrolytes YES
Free-water phase properties method STEAM-TA
Water solubility method 3
Specified pressure [bar] 25
Specified temperature [K] 330
Specified heat duty [cal/sec]
Reactor volume [l]
Reactor residence time [hr] 0.5
Phase volume [l]
Phase volume frac
Phase residence time [hr]
Change grid orientation

REACTOR VOLUME = 8827.87 L = 8.827 m^3



EO Model components	
Outlet temperature [C]	56.85
Calculated heat duty [cal/sec]	-11956.1
Net heat duty [cal/sec]	-11956.1
Reactor volume [l]	8827.87
Vapor phase volume	
Liquid phase volume [l]	8827.87
Liquid 1 phase volume	
Salt phase volume	
Condensed phase volume [l]	8827.87
Reactor residence time [hr]	0.499998
Vapor phase residence time	
Condensed phase residence time [hr]	0.499998
Total feed stream CO2e flow [kg/hr]	50.2186
Total product stream CO2e flow [kg/hr]	41.4075
Net stream CO2e production [kg/hr]	-8.81117



EO Model components	
Outlet temperature [C]	56.85
Calculated heat duty [cal/sec]	-11956.1
Net heat duty [cal/sec]	-11956.1
Reactor volume [l]	8827.87
Vapor phase volume	
Liquid phase volume [l]	8827.87
Liquid 1 phase volume	
Salt phase volume	
Condensed phase volume [l]	8827.87
Reactor residence time [hr]	0.499998
Vapor phase residence time	
Condensed phase residence time [hr]	0.499998
Total feed stream CO2e flow [kg/hr]	50.2186
Total product stream CO2e flow [kg/hr]	41.4075
Net stream CO2e production [kg/hr]	-8.81117